

Stanford Background Dataset 语义分割实验报告

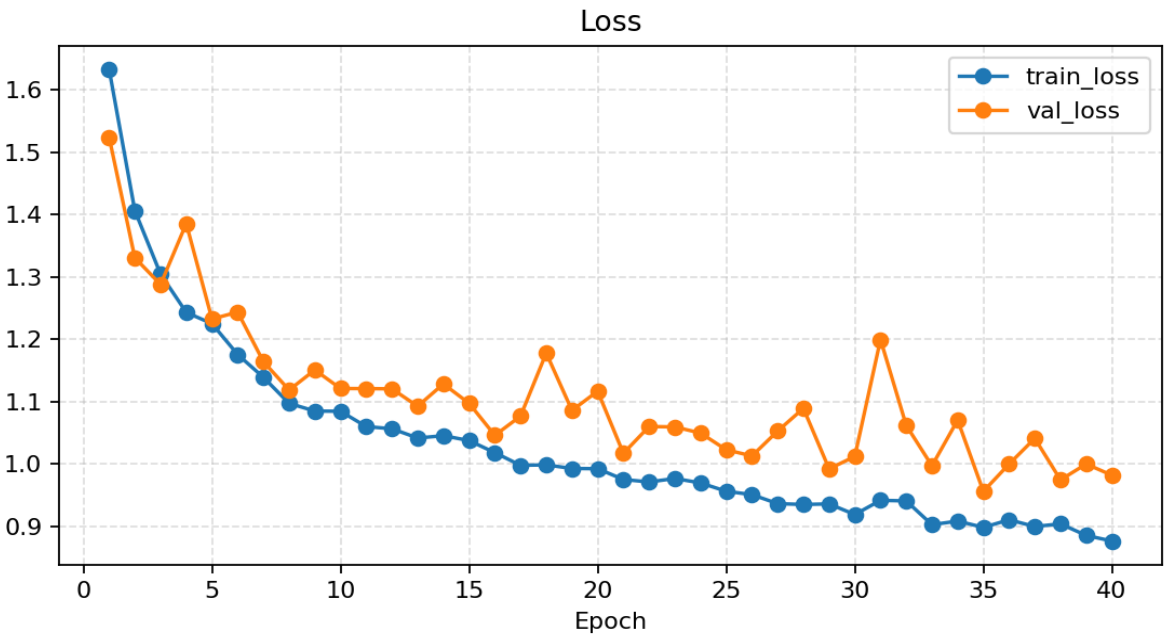
本报告汇总 U-Net 在 Cross-Entropy、Dice Loss 与组合损失三种配置下的训练和测试结果。

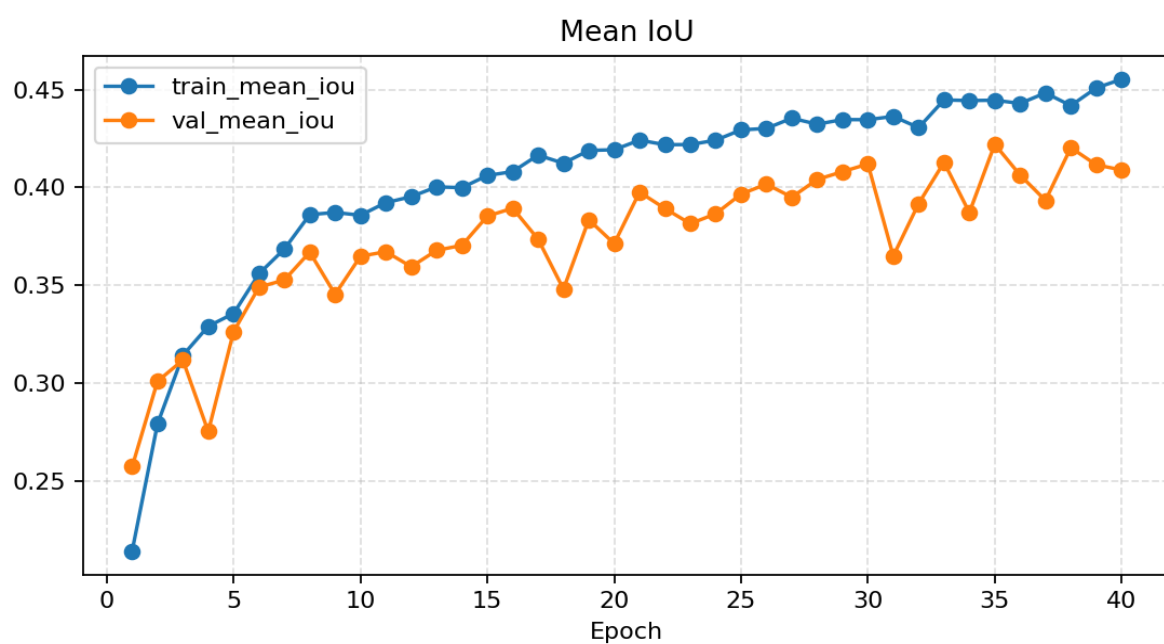
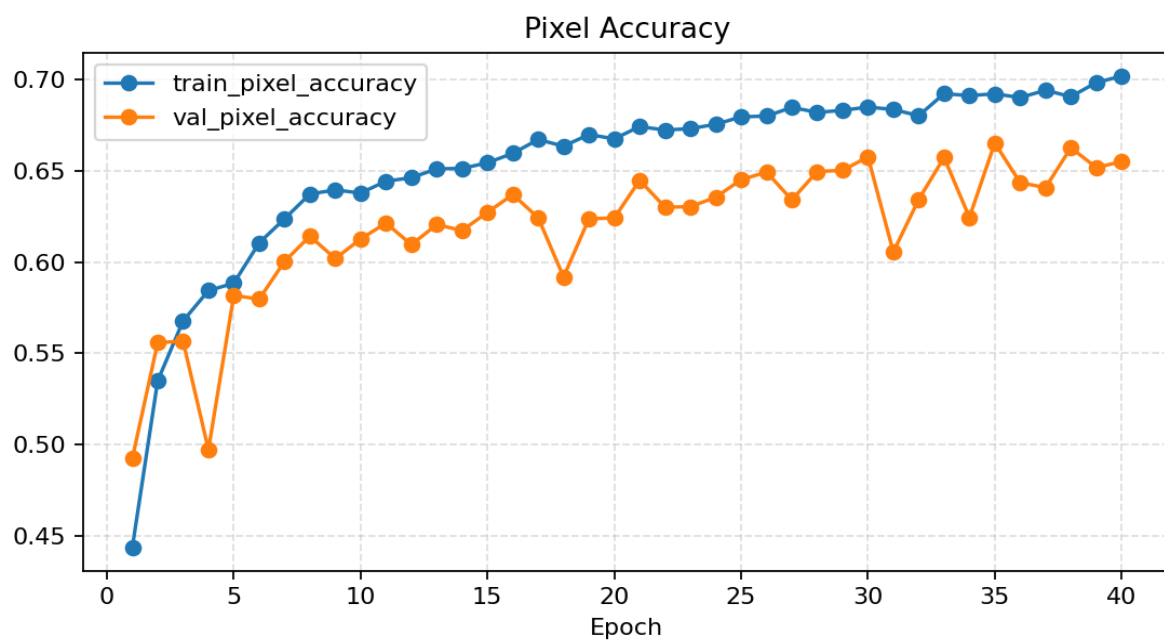
数据集与模型

数据集使用 Stanford Background Dataset, 包含 sky、tree、road、grass、water、building、mountain 和 foreground object 共 8 类; 负数或越界标签作为 unknown 像素, 在损失和评估中忽略。模型为从零训练的经典 U-Net, 包含编码器、解码器和同尺度 skip connection 拼接。

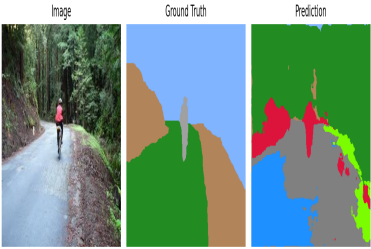
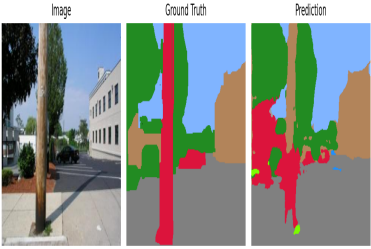
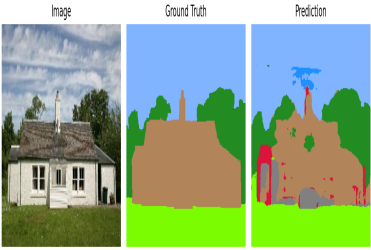
实验	损失函数	Batch	学习率	Epoch	验证 mIoU	测试 Acc	测试 mIoU
unet_ce	ce	8	0.001	40	0.4220	0.6545	0.4187
unet_ce_dice	ce_dice	8	0.001	40	0.4232	0.6403	0.4150
unet_dice	dice	8	0.001	40	0.4136	0.6174	0.4022

unet_ce 实验曲线

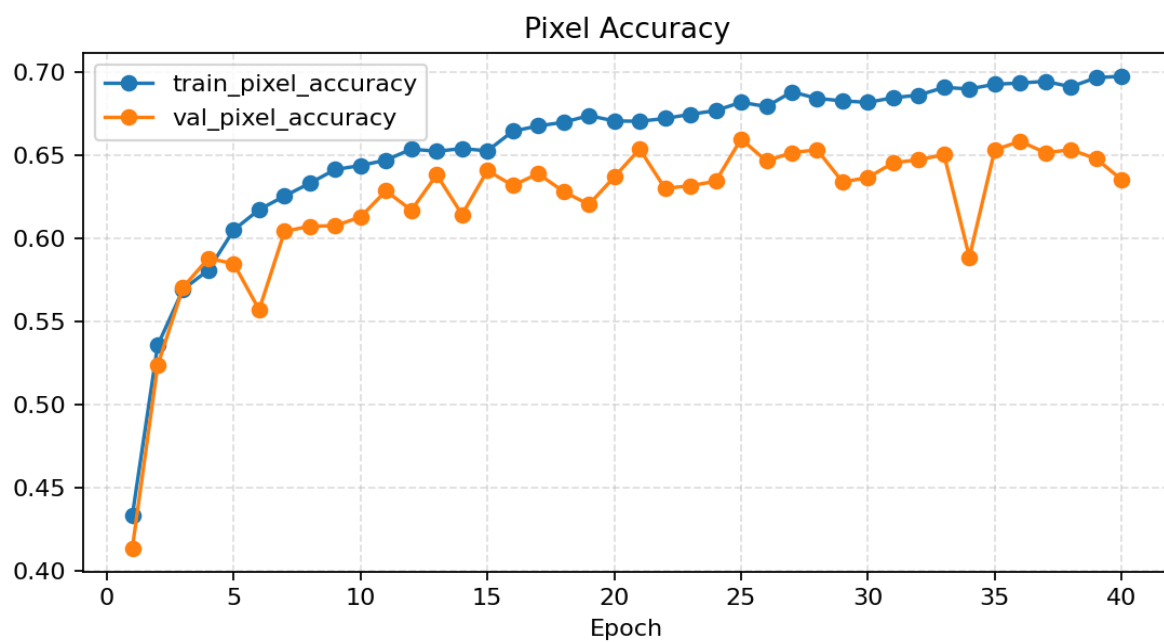
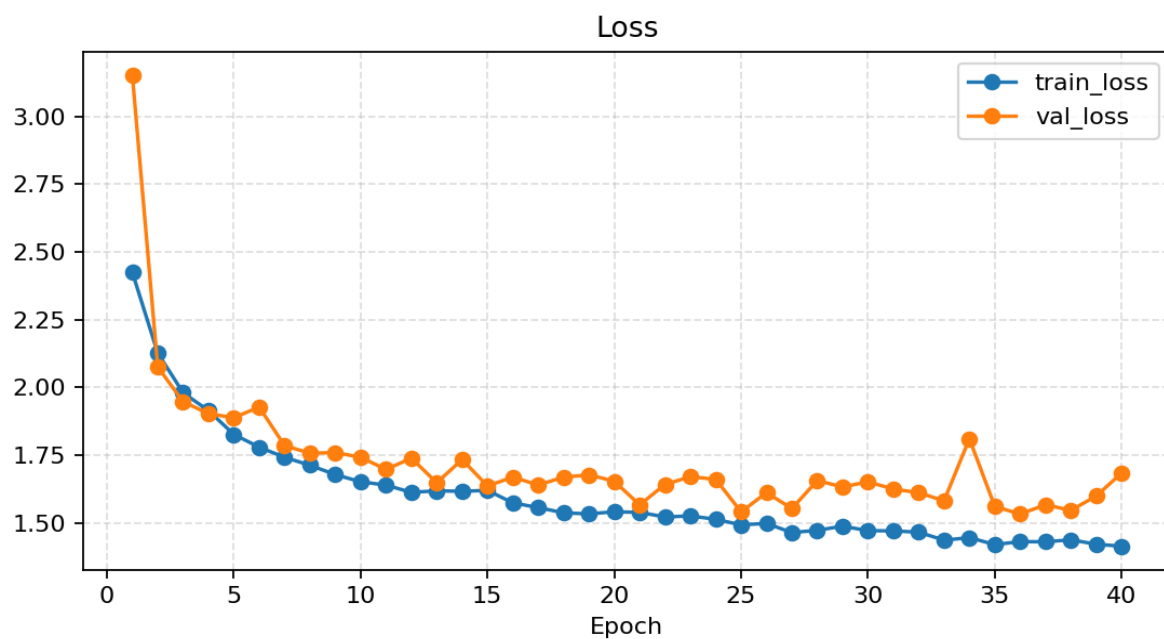


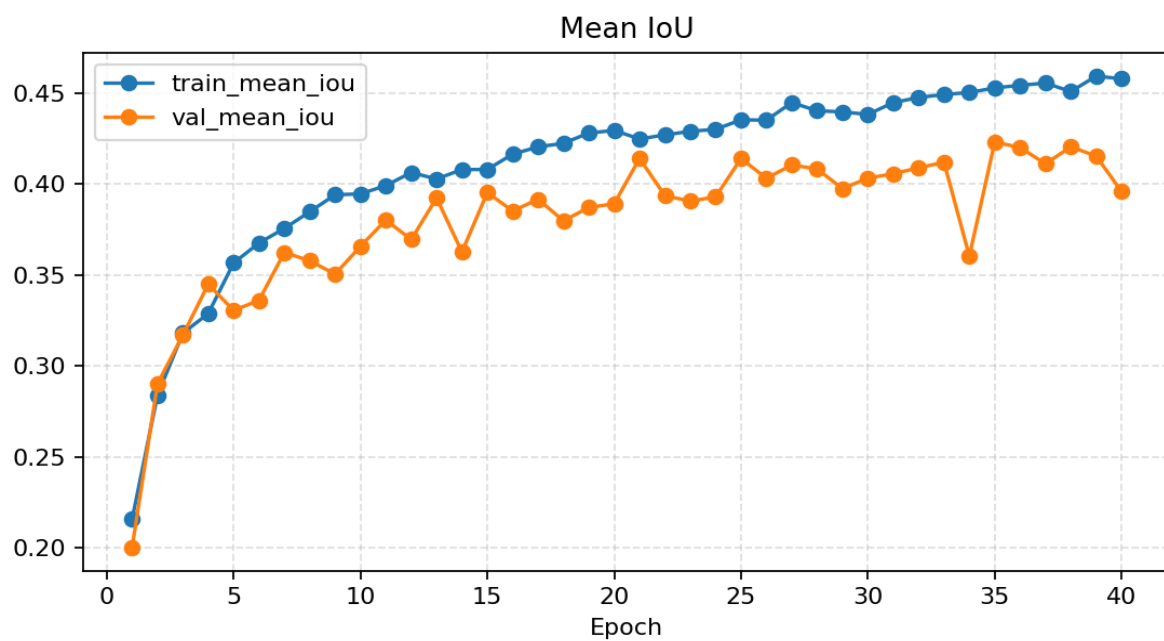


测试集可视化

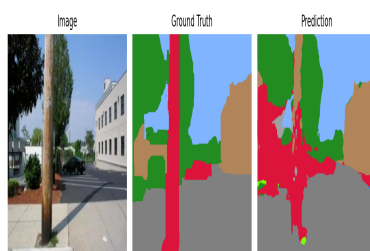
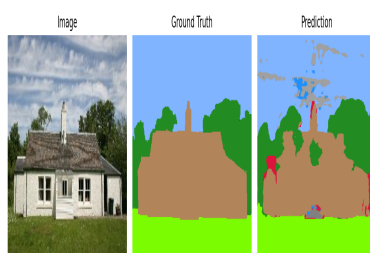


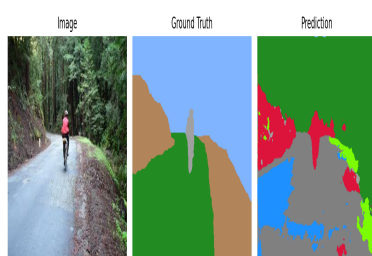
unet_ce_dice 实验曲线



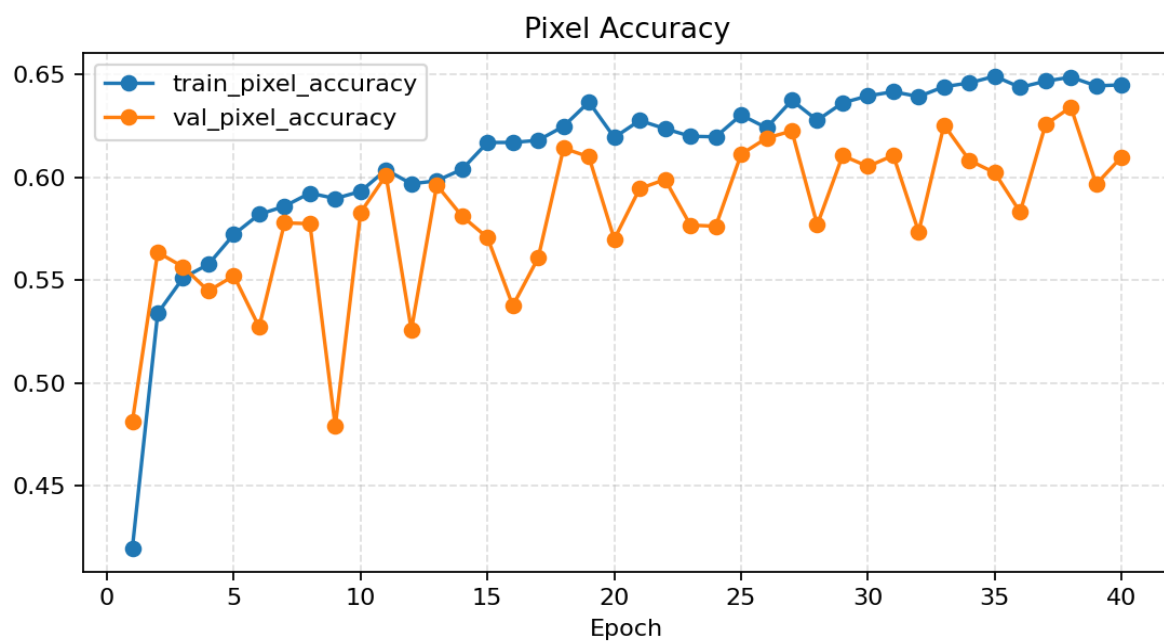
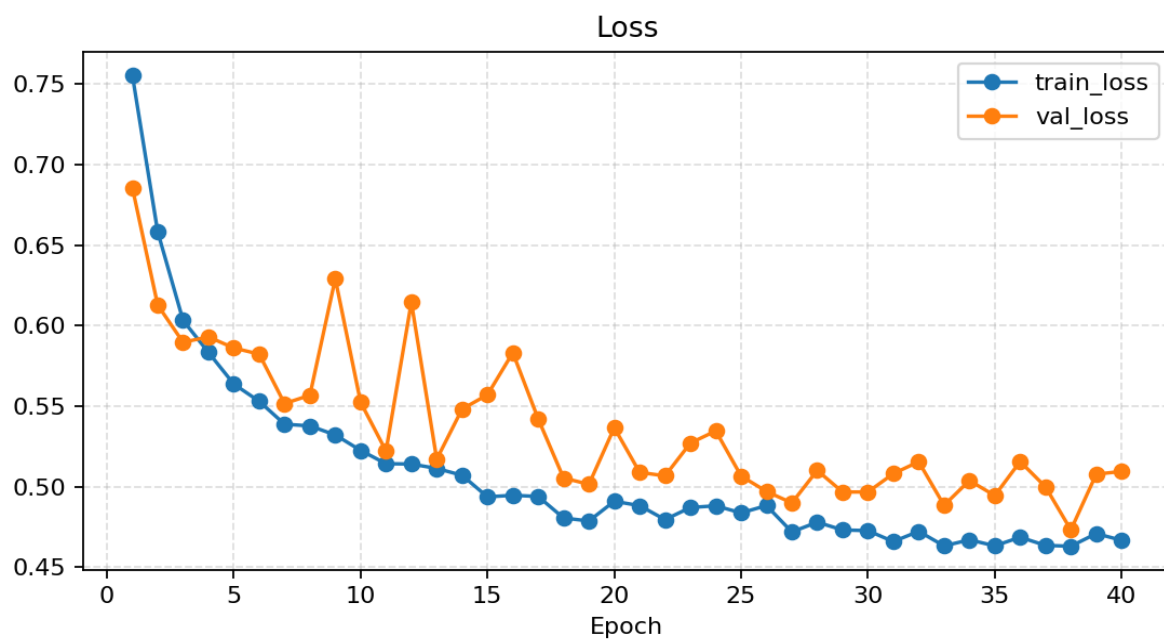


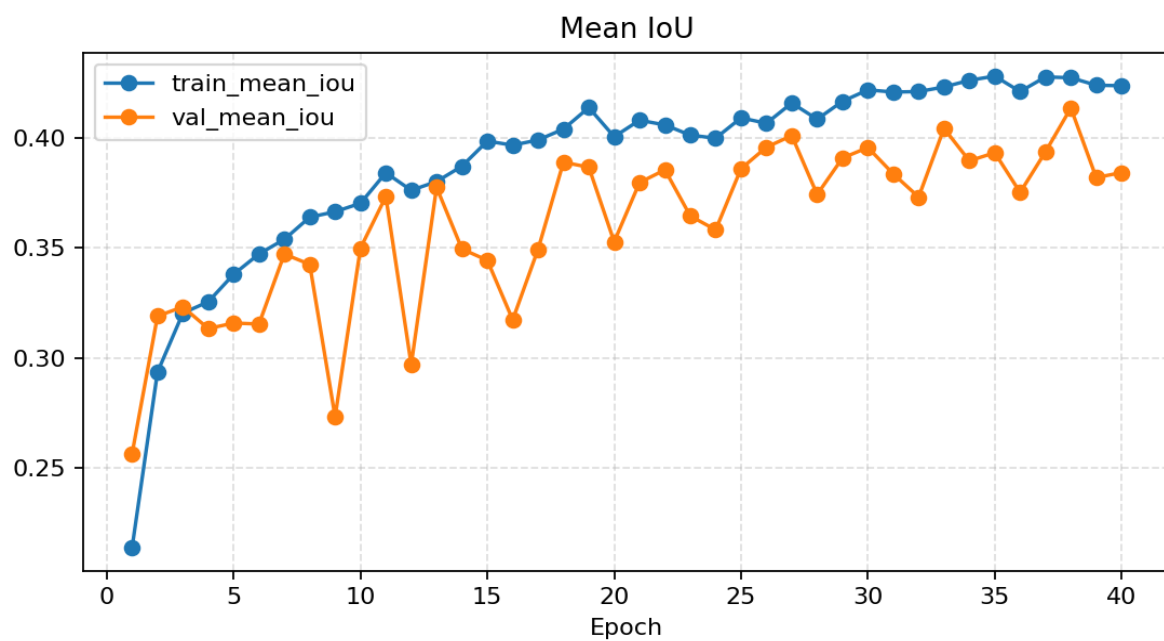
测试集可视化



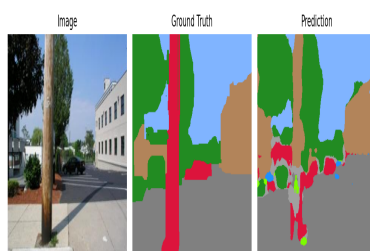
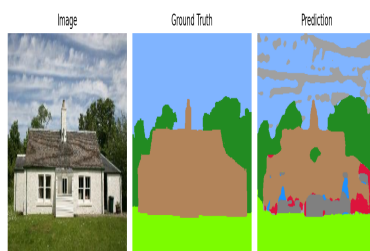


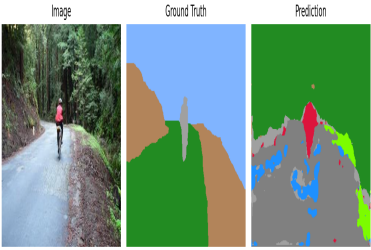
unet_dice 实验曲线





测试集可视化





结论

Cross-Entropy 通常收敛稳定, Dice
更适合缓解类别不均衡, 组合损失往往在精度与区域重叠之间取得更均衡的表现。